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SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY

Autonomous Institution

Yamnampet, Ghatkesar, Hyderabad – 501 301.

FIRST MID EXAMINATION, September- 2025

Course: B.Tech IV Year I SEM **Branch:** CSE/IT/AIML/DS/IoT

Subject: Cloud Computing (9FC17) **Time** : 2 Hours

Date : 25 .09 .2025 (FN) **Max Marks:** 30

Part – A

(Compulsory Short answers) 6X 2 = 12 Marks

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-I	Define Cloud Computing? Explain essential characteristics of cloud computing	L1	CO I	2
2	From Unit-I	Define Virtualization? What is the purpose of Hypervisor in Cloud Computing?	L1	CO I	2
3	From Unit-II	Briefly explain cloud application services.	L1	CO 1I	2
4	From Unit-II	Write short notes on Open stack Open Source Private Cloud Software.	L3	CO II	2
5	From Unit-III	What are the pros and cons of cloud data storage approaches?	L1	CO III	2
6	From Unit-III	Write short notes on REST?	L1	CO III	2

Part – B

Answer any THREE out of the FOUR questions 3X 6 = 18 Marks

S.No	Coverage		BCLL	CO(s)	Marks
7	From Unit-I	Discuss in detail the Cloud Computing service and Deployment models.	L2	CO I	6
8	From Unit-II	Explain in detail the computing and Storage services offered by different types of Cloud Service Providers.	L2	CO II	6
9	From Unit-III	Illustrate in detail about cloud application design methodologies.	L4	CO III	6
10 a)	From Unit-I	Briefly explain Map Reduce with suitable example.	L4	CO I	2
b)	From Unit-II	Explain the Analytics services offered by different types of Cloud Service Providers.	L3	CO II	2
c)	From Unit-III	Discuss in detail Reference Architectures for Cloud Applications.	L2	CO III	2

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Yamnampet, Ghatkesar, Hyderabad – 501 301.

FIRST MID EXAMINATION, September- 2025

Course: B.Tech IV Year I SEM

Subject: Agile Software Development -9FC16

Date : 26 .09 .2025 (AN)

Branch: CSE & DS

Time : 2 Hours

Max Marks: 30

Part – A

(Compulsory Short answers) 6X 2 = 12 Marks

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-I	Define Agile?	L1	CO1	2
2	From Unit-I	Name some agile models?	L2	CO1	2
3	From Unit-II	What are the values of ASD?	L1	CO2	2
4	From Unit-II	Define Pair Programming?	L1	CO2	2
5	From Unit-III	Write Phases in XP Project?	L2	CO3	2
6	From Unit-III	Define refactoring?	L1	CO3	2

Part – B

Answer any THREE out of the FOUR questions 3X 6 = 18 Marks

S.No	Coverage		BCLL	CO(s)	Marks
7	From Unit-I	What is Agile Manifesto? Explain in detail?	L2	CO1	6
8	From Unit-II	Explain XP life cycle with neat diagram?	L2	CO2	6
9	From Unit-III	Explain how iterative development is practiced in both AM and XP?	L3	CO3	6
10 a)	From Unit-I	Write about SCRUM?	L1	CO1	2
b)	From Unit-II	Name some Practices of XP?	L2	CO2	2
c)	From Unit-III	Define the Fit?	L1	CO3	2

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FIRST MID EXAMINATION, September- 2025

Course: B.Tech IV Year I SEM **Branch:** MECH / CSE/ IT/CS/AIML/DS/IOT

Subject: 9ZC23- Advanced Entrepreneurship **Time** : 2 Hours

Date : 25 .09 .2025 (AN) **Max Marks:** 30

Part – A

(Compulsory Short answers) 6X 2 = 12 Marks

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-I	Define the concept of Pivoting.	L1	Co1	2m
2	From Unit-I	What are the functions of entrepreneurs?	L1	Co1	2m
3	From Unit-II	What is financial planning?	L1	C02	2m
4	From Unit-II	Write the secondary sources of revenue for start-up?	L1	C02	2m
5	From Unit-III	Elucidate the importance of crowd funding	L6	C02	2m
6	From Unit-III	Write the advantages of boot strapping	L1	C02	2m

Part – B

Answer any THREE out of the FOUR questions **3X 6 = 18 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
7	From Unit-I	Illustrate the various types of Business models with suitable examples.	L4	CO1	6M
8	From Unit-II	Explain business planning and types of business planning?	L2	CO2	6M
9	From Unit-III	Write the various forms of funding options for an entrepreneur?	L1	CO3	6M
10 a)	From Unit-I	Mention the different skills of entrepreneurship	L1	CO1	2M
b)	From Unit-II	Write the secondary sources of revenue for start-up?	L1	CO2	2M
c)	From Unit-III	Write the importance of building A-startup team	L1	CO3	2M

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Yamnampet, Ghatkesar, Hyderabad – 501 301.

FIRST MID EXAMINATION, September - 2025

Course: B.Tech IV Year I SEM
Branch: CSE/IT
Subject: Cryptography and Network Security-**9EC09**
Time : 2 Hours
Date : 23 .09 .2025 (FN)
Max Marks: 30

Part – A

(Compulsory Short answers) **6X 2 = 12 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-I	Explain about security goals?	L2	CO1	2
2	From Unit-I	Name the types of security mechanisms?	L1	CO1	2
3	From Unit-II	List any two differences between DES and AES algorithms	L1	CO2	2
4	From Unit-II	What are the types of cipher block modes of operation?	L1	CO2	2
5	From Unit-III	Define for digital signature?	L2	CO3	2
6	From Unit-III	Explain the concept of Kerberos?	L2	CO3	2

Part – B

Answer any THREE out of the FOUR questions **3X 6 = 18 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
7	From Unit-I	Describe briefly about types of security Services.	L2	CO1	6
8	From Unit-II	Explain RSA Algorithm with an example?	L2	CO2	6
9	From Unit-III	Discuss in detail about HMAC algorithm?	L6	CO3	6
10 a)	From Unit-I	Mention the types of security Attacks?	L2	CO1	2
b)	From Unit-II	Differentiate double DES and TDES?	L4	CO2	2
c)	From Unit-III	Define S/MIME?	L1	CO3	2

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FIRST MID EXAMINATION, September- 2025

Course: B.Tech IV Year I SEM
Subject: Linux Programming (9EC08)
Date : 23-09-2025 (AN)

Branch: CSE / IT
Time : 2 Hours
Max Marks: 30

Part – A

(Compulsory Short answers) **6X 2 = 12 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-I	Explain the purpose of the **head** command with example.	L2	CO1	2
2	From Unit-I	List any four disk utilities in Linux.	L1	CO1	2
3	From Unit-II	Define command substitution with an example.	L2	CO2	2
4	From Unit-II	Write the syntax of **for** loop in bash.	L2	CO2	2
5	From Unit-III	What is meant by stream error in C?	L1	CO2	2
6	From Unit-III	Define absolute and relative file paths.	L2	CO2	2

Part – B

Answer any THREE out of the FOUR questions **3X 6 = 18 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
7 a)	From Unit-I	Explain the working of **grep** command with an example.	L2	CO1	3
7 b)		Discuss the purpose of backup utilities in Linux.	L2	CO1	3
8 a)	From Unit-II	Explain the purpose of the **case** statement in bash scripting with an example.	L3	CO2	3
8 b)		Discuss arithmetic evaluation using **expr** in bash with an example.	L3	CO2	3
9 a)	From Unit-III	Explain formatted I/O functions in C with example.	L2	CO2	3
b)		Discuss the role of **read ()** and **write ()** system calls.	L3	CO2	3
10 a)	From Unit-I	Explain the usage of addresses in sed.	L3	CO2	2
b)	From Unit-II	Discuss quoting mechanisms in bash scripting.	L2	CO2	2
c)	From Unit-III	Write two differences between system calls and library functions.	L4	CO2	2

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FIRST MID EXAMINATION, September- 2025

Course: B.Tech IV Year I SEM
Branch : CSE/IT
Subject: 9EC07-Automata Theory and Compiler Design
Time : 2 Hours
Date : 26 .09 .2025 (FN)
Max Marks: 30

Part – A

(Compulsory Short answers) **6X 2 = 12 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-I	Design a DFA for the language over Σ={a,b} that accepts the following i) All strings with exactly one a ii) All strings with atleast one a	L3	CO1	2
2	From Unit-I	Define finite state machine and finite automata	L1	CO1	2
3	From Unit-II	What is meant by Ambiguous grammar? Show that the following grammar is ambiguous S→aSb/aaSb/ε	L2	CO2	2
4	From Unit-II	Find the simplified regular expression for the following regular expression r (r*r + r*) + r*.	L4	CO2	2
5	From Unit-III	Give the formal definition Turing Machine	L1	CO3	2
6	From Unit-III	Convert the following grammar into PDA S→aSb S→ab	L3	CO3	2

Part – B

Answer any <u>THREE</u> out of the FOUR questions **3X 6 = 18 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
7	Unit-I	(i) Design DFA which accepts all strings over {a,b} starting with '10' and ending with 01' (ii) Design DFA which accepts all strings containing both even number of 0's and even number of 1's.	L6	CO1	3 3
8	Unit-II	(i) Construct finite automata for the regular expression (1*0 +10*). (ii) Convert the following grammar into CNF $S \to ABa$ $A \to aab$ $B \to Ac$	L3 L4	CO2	3 3
9	Unit-III	Write the procedure to convert CFG to PDA and also convert the following CFG to PDA. S --> B | aAA A → aBB | a B → bBB | A C → a	L6	CO3	6
10 a)	Unit-I	Explain about Chomsky Hierarchy of languages	L1	CO1	2
b)	Unit-II	Define Greiback Normal Form	L1	CO2	2
c)	Unit-III	Define PushDown automata .Obtain a PDA to accept strings of balanced parantheses	L1	CO3	2

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SECOND MID EXAMINATION, December 2025

Course: B.Tech IV Year I SEM **Branch:** CSE/IT/AIML/DS/IOT
Subject: Cloud Computing - 9FC17 **Time** : 2 Hours
Date : 03 .12 .2025 (FN) **Max Marks:** 30

Part – A
(Compulsory Short answers) **6X 2 = 12 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-IV	How is IoT used in smart agriculture?	L1	CO4	2
2	From Unit-IV	What is latency in distributed systems?	L1	CO4	2
3	From Unit-V	What is the role of pre-processing in image processing apps?	L1	CO5	2
4	From Unit-V	Mention one Python library used to implement Map Reduce.?	L1	CO5	2
5	From Unit-VI	What is the purpose of a Cloud Security Architecture?	L1	CO6	2
6	From Unit-VI	Mention any two techniques used to secure cloud data.	L1	CO6	2

Part – B
Answer any THREE out of the FOUR questions **3X 6 = 18 Marks**

S. No	Coverage		BCLL.	CO(s)	Marks
7	From Unit-IV	Describe the key enabling technologies for IoT	L2	CO4	6
8	From Unit-V	Explain the architecture of a cloud image processing system using Python and cloud storage?	L2	C05	6
9	From Unit-VI	Describe the key components of Cloud Security Architecture (CSA).	L2	CO6	6
10 a)	From Unit-IV	Give two examples of IoT applications in everyday life?	L1	CO4	2
b)	From Unit-V	What is distributed data processing?	L1	CO5	2
c)	From Unit-VI	State any two common authentication methods used in cloud systems.	L3	CO6	2

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SECOND MID EXAMINATION, December 2025

Course: B.Tech IV Year I SEM **Branch:** CSE/DS
Subject: AGILE SOFTWARE DEVELOPMENT (9FC16) **Time :** 2 Hours
Date : 04 .11 .2025(AN) **Max Marks:** 30

Part – A
(Compulsory Short answers) **6X 2 = 12 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-IV	Define Feature-Driven Development (FDD).	L2	CO4	2
2	From Unit-IV	What is meant by architecture-centric development.	L1	CO4	2
3	From Unit-V	Write any two benefits of integrating Agile with PRINCE2.	L1	CO5	2
4	From Unit-V	Give an example of an Agile modeling practice.	L2	CO5	2
5	From Unit-VI	What is the failed project syndrome.	L1	CO6	2
6	From Unit-VI	State any two challenges in converting a traditional team to Agile.	L2	CO6	2

Part – B
Answer any THREE out of the FOUR questions **3X 6 = 18 Marks**

S. No	Coverage		BCLL	CO(s)	Marks
7	From Unit-IV	Explain the five major processes of Feature-Driven Development (FDD). P64-Y	L4	CO4	6
8	From Unit-V	Discuss how Agile principles improve RUP-based projects.	L6	CO5	6
9	From Unit-VI	Discuss the major obstacles to Agile software development with examples. No commitm lust not al	L6	CO6	6
10 a)	From Unit-IV	List and explain any three key characteristics of FDD.	L2	CO4	2
b)	From Unit-V	Write a short note on any three tools used in Agile Development.	L1	CO5	2
c)	From Unit-VI	Why does poor training affect Agile projects. less fault of dev	L1	CO6	2

version control
Git. Git hub

Communict teams

project mangmt tool
trello
Jira

Customer-centric-d
Quality devp
clear communct
new for roles.

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SECOND MID EXAMINATION, December 2025

Course: B. Tech IV Year I SEM **Branch:** ME/CSE/IT/CS/AIMLI/DS/IOT
Subject: ADVANCED ENTREPRENEURSHIP – 9ZC23 **Time:** 2 Hours
Date : 03.12 .2025 (AN) **Max Marks:** 30

Part – A
(Compulsory Short answers) **6X 2 = 12 Marks**

S. No	Coverage		BCLL	CO(s)	Marks
1	From Unit-IV	Define branding and draw the venture's golden circle	L1	CO-IV	2
2	From Unit-IV	What are the metrics for customer acquisition and also List out the features of digital collaboration	L2	CO-IV	2
3	From Unit-V	Define customer churn rate and list out any two financial metrics?	L1	CO-V	2
4	From Unit-V	What are the steps we have to remember while selecting company name	L3	CO-V	2
5	From Unit-VI	What is the role of mentors and Types of strategies for effective mentor mentee relationship?	L1	CO-VI	2
6	From Unit-VI	Define Capstone project and list out any three steps under legal compliance	L2	CO-VI	2

Part – B
Answer any THREE out of the FOUR questions **3X 6 = 18 Marks**

S. No	Coverage		BCLL	CO(s)	Marks
7	From Unit-IV	Write the role of technology and online platforms for expanding the reach of start-ups?	L2	CO-IV	6
8	From Unit-V	What do you understand by financial metrics? & Explain in brief?	L3	CO-V	6
9	From Unit-VI	What are the main contents for final presentation of the start-up project to the investors?	L1	CO-VI	6
10 a)	From Unit-IV	Distinguish between Mentor and Advisor	L1	CO-IV	2
b)	From Unit-V	Elaborate ARPU.	L2	CO-V	2
c)	From Unit-VI	Exemplify the customer life time value.	L3	CO-VI	2

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SECOND MID EXAMINATION, December 2025

Course: B.Tech IV Year I SEM **Branch:** CSE/IT

Subject: Cryptography and Network Security -9EC09 **Time:** 2 Hours

Date : 01 .12 .2025 (FN) **Max Marks:** 30

Part – A

(Compulsory Short answers) **6X 2 = 12 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-IV	Explain about IP Security.	BL2	CO	2M
2	From Unit-IV	What is tunnel mode and transport mode in IPSec?	BL1	CO	2M
3	From Unit-V	Define SSL and TLS.	BL1	CO	2M
4	From Unit-V	What is Digital Certificate?	BL1	CO	2M
5	From Unit-VI	Explain about trusted System.	BL2	CO	2M
6	From Unit-VI	Compare and contrast between IDS and IPS.	BL2	CO	2M

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Part – B

Answer any THREE out of the FOUR questions **3X 6 = 18 Marks**

S. No	Coverage		BCLL	CO(s)	Marks
7	From Unit-IV	Describe Authentication Header (AH) in detail.	BL1	CO4	6M
8	From Unit-V	Explain SSL architecture and working of SSL protocols.	BL2	CO5	6M
9	From Unit-VI	Discuss Intrusion detection System and classify them.	BL1	CO6	6M
10 a)	From Unit-IV	Demonstrate about Security Association(SA).	BL2	CO4	2M
b)	From Unit-V	Explain various web security threats.	BL2	CO5	2M
c)	From Unit-VI	Discuss firewall design principles.	BL1	CO6	2M

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SECOND MID EXAMINATION, December 2025

Course: B.Tech IV Year I SEM

Subject: LINUX PROGRAMMING(9EC08)

Date : 01.12.2025 (AN)

Branch: CSE/IT

Time : 2 Hours

Max Marks: 30

Part – A

(Compulsory Short answers) **6X 2 = 12 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-IV	Show difference between dup() and dup2().	1	4	2
2	From Unit-IV	Interpret signal( ) prototype.	2	4	2
3	From Unit-V	State the need for IPC.	1	5	2
4	From Unit-V	Recall the msgctl( ) functionalities.	1	5	2
5	From Unit-VI	List operations on shared memory.	1	6	2
6	From Unit-VI	Specify the applications for message queue.	1	6	2

Part – B

Answer any THREE out of the FOUR questions **3X 6 = 18 Marks**

S. No	Coverage		BCLL	CO(s)	Marks
7	From Unit-IV	a) Illustrate how to create a process?	4	4	3
		b) Distinguish between reliable and unreliable signals.	2	4	3
8	From Unit-V	a) Make use of mkfifo( ) to create a FIFO and list about its associated functions.	3	5	3
		b) Criticize issues associated with pipes.	5	5	3
9	From Unit-VI	a) Write about semget( ).	6	6	3
		b) Explain shmctl( )with possible cmd strings.	2	6	3
10 a)	From Unit-IV	a) Tell about Zombie process.	1	4	2
b)	From Unit-V	b) Classify UNIX SYSTEM V IPC mechanisms.	2	5	2
c)	From Unit-VI	c) Show the applications shared memory mechanism.	1	6	2

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THIRD MID EXAMINATION, January-2026

Course: B.Tech IV Year I SEM **Branch:** CSE, IT
Subject: 9EC08-Linux Programming **Time** : 2 Hours
Date : 03 .01 .2026 (AN) **Max Marks:** 30

Part – A
(Compulsory Short answers) **6X 2 = 12 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
1	Unit-I	Write short notes on ***expr*** and ***eval***	2	1	2
2	Unit-II	Explain command execution and its types	2	1	2
3	Unit-III	What are the Library functions.	1	2	2
4	Unit-IV	Explain about interrupted system calls	2	2	2
5	Unit-V	Define pipe?	1	3	2
6	Unit-VI	Explain how to attach and detach a shared-memory segment.	2	4	2

Part – B
Answer any THREE out of the SIX questions **3X 6 = 18 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
7	Unit-I	Explain the any four network related commands.	2	1	6
8	Unit-II	Define Shell and explain its Responsibilities.	2	1	6
9	Unit-III	Differentiate between the following terms: (a) getc( ) Vs fgetc( ) (b) printf( ) Vs fprint( ) (c) scanf( ) Vs fscanf( )	2	2	6
10	Unit-IV	Explain about vfork, exit, wait and exec functions?	2	2	6
11	Unit-V	Illustrate mkfifo( ) system call with an example.	3	3	6
12	Unit-VI	Explain the kernel data structure for semaphores with a neat diagram. Also explain the APIs associated for semaphores.	2	4	6

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I/O redirect
process exit.
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SECOND MID EXAMINATION, December 2025

Course: B.Tech IV Year I SEM **Branch:** CSE/IT

Subject: AUTOMATA THEORY and COMPILER DESIGN - 9EC07 **Time:** 2 Hrs

Date : 04.12.2025 (FN) **Max Marks:** 30

Part – A

(Compulsory Short answers) 6X 2 = 12 Marks

S.No	Coverage		BCLL	CO(s)	Marks
1	Unit-IV	What is the difference between a compiler phase and a compiler pass?	L1	CO3	2
2	Unit-IV	What is Recursive Descent Parsing? Give an example	L1	CO3	2
3	Unit-V	What is a handle in bottom-up parsing? Give an example.	L1	CO4	2
4	Unit-V	What is the role of an operator-precedence parser?	L1	CO4	2
5	Unit-VI	What is the purpose of a symbol table? Give an example.	L1	CO5	2
6	Unit-VI	Define peephole optimization with an example.	L1	CO5	2

Part – B

Answer any <u>THREE</u> out of the FOUR questions 3X 6 = 18 Marks

S. No	Coverage		BCLL	CO(s)	Marks
7	Unit-IV	Design LL(1) Parsing Table for the following grammar and Parse the string "id+id*id". E-> E + T | T T-> T * F | F F -> (E) / id	L4	CO3	6
8	Unit-V	Design SLR Parsing Table for the following grammar and Parse the String "id*id+id". E-> E + T | T T-> T * F | F F -> (E) / id	L4	CO4	6
9	Unit-VI	Write quadruples, triples and indirect triples for the expression: a = – c * b + – c * b	L2	CO5	6
10 a)	Unit-IV	Write the output for all the phases of compiler for the following expression: position: = initial + rate * 60	L2	CO3	2
b)	Unit-V	Consider below grammar and Perform shift reduce parsing of the input string “id-id*id”. E-> E-E E-> E*E E-> id	L3	CO4	2
c)	From Unit-VI	Draw the syntax tree and DAG for the following expression: a = b * – c + b * – c	L3	CO5	2

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Sreenidhi Institute of Science and Technology

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Code No:9FC17 **Date:27-Dec-2025 (AN)**

B.Tech IV-Year I-Semester External Examination, December-2025 (Regular)

CLOUD COMPUTING (PE-V) (CSE,IT,AIML,DS & IOT)

Time: 3 Hours **Max.Marks:60**

Note: *a) No additional answer sheets will be provided.*
b) All sub-parts of a question must be answered at one place only, otherwise it will not be valued.
c) Missing data can be assumed suitably.

Bloom's Cognitive Levels of Learning (BCLL)

Remember	L1	Apply	L3	Evaluate	L5
Understand	L2	Analyze	L4	Create	L6

Part - A **Max.Marks: 6x2=12**

ANSWER ALL QUESTIONS, EACH QUESTION CARRIES 2 MARKS.

		BCLL	CO(s)	Marks
1	Discuss the importance of scalability in cloud.	L2	CO1	[2M]
2	Define compute service with an example. RFID	L1	CO2	[2M]
3	Discuss any two data storage approaches in cloud applications.	L2	CO3	[2M]
4	List the enabling technologies of IoT.	L1	CO4	[2M]
5	Write a short note on any two Python libraries used in cloud application development.	L1	CO5	[2M]
6	Define data security in cloud environments.	L1	CO6	[2M]

Part – B **Max.Marks: 6x8=48**

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 8 MARKS.

		BCLL	CO(s)	Marks
7.	Analyze the differences between IaaS, PaaS, and SaaS with suitable examples.	L4	CO1	[8M]
	OR			
8	List and explain the various types of cloud deployment models with suitable examples.	L1	CO1	[8M]
9.	Describe the role of Identity and Access Management in cloud security with example.	L1	CO2	[8M]
	OR			
10	Explain in detail about compute, storage, and database services offered by cloud providers.	L2	CO2	[8M]
11	Evaluate the effectiveness of using MapReduce for large-scale data processing.	L5	CO3	[8M]
	OR			
12	List and explain the various design considerations for building cloud applications.	L6	CO3	[8M]
13	Evaluate the security issues in IoT-cloud systems and propose the mitigation strategies.	L5	CO4	[8M]
	OR			
14	Analyze performance challenges of distributed IoT systems connected to cloud environments.	L4	CO4	[8M]
15	Describe the steps in creating a Python based MapReduce application.	L1	CO5	[8M]
	OR			
16	Apply Python APIs to build a document storage cloud application.	L3	CO5	[8M]
17	Explain about cloud security architecture (CSA) with neat diagram.	L2	CO6	[8M]
	OR			
18	Discuss how cloud security is applied in industry, healthcare, and education with examples.	L2	CO6	[8M]

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Sreenidhi Institute of Science and Technology
(An Autonomous Institution)

Regulations: A22

Code No:9FC16

Date:24-Dec-2025 (AN)

B.Tech IV-Year I-Semester External Examination, December-2025 (Regular)

AGILE SOFTWARE DEVELOPMENT (CSE & DS)

Time: 3 Hours **Max.Marks:60**

Note: a) *No additional answer sheets will be provided.*
b) *All sub-parts of a question must be answered at one place only, otherwise it will not be valued.*
c) *Missing data can be assumed suitably.*

Bloom's Cognitive Levels of Learning (BCLL)

Remember	L1	Apply	L3	Evaluate	L5
Understand	L2	Analyze	L4	Create	L6

Part - A

Max.Marks: 6x2=12

ANSWER ALL QUESTIONS, EACH QUESTION CARRIES 2 MARKS.

		BCLL	CO(s)	Marks
1	Define Agile Manifesto and list its values.	L1	CO1	[2M]
2	Explain the concept of pair programming.	L2	CO2	[2M]
3	Compare Agile Modeling and XP	L2	CO3	[2M]
4	What is Feature-Driven Development (FDD)?	L1	CO4	[2M]
5	Mention tools that support Agile Development	L1	CO5	[2M]
6	List any obstacles to Agile Software Development	L1	CO5	[2M]

Part – B

Max.Marks: 6x8=48

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 8 MARKS.

e-Com Mobile app

		BCLL	CO(s)	Marks
7	Discuss in detail the Agile Methods: XP, DSDM and SCRUM with suitable examples.	L5	CO1	[8M]
	OR			
8	Describe XP, DSDM, SCRUM, and FDD methodologies and compare their key characteristics.	L4	CO1	[8M]
9.	Explain the Twelve XP Practices and how they support real-time project development	L6	CO2	[8M]
	OR			
10	Demonstrate how Test-First Development and Pair Programming can be implemented in a project environment.	L3	CO2	[8M]
11	Describe the relationship between Agile Modeling and Planning XP Projects with a neat diagram.	L4	CO3	[8M]
	OR			
12	Describe the common and modeling-specific practices of Agile Modeling with examples.	L4	CO3	[8M]
13	Explain the different phases of Feature-Driven Development and how it helps regain control over project development	L4	CO4	[8M]
	OR			
14	Explain how FDD uses incremental software development to manage large projects. Give examples.	L4	CO4	[8M]
15	Describe Agile Methods with RUP and PRINCE2 and evaluate tools available to support Agile Development.	L5	CO5	[8M]
	OR			

16	Explain how Eclipse IDE supports Agile practices such as refactoring and unit testing.	L4	CO5	[8M]
17	Examine managerial, contractual and project-based obstacles that affect Agile SDLC	L3	CO5	[8M]
	OR			
18	Evaluate the importance of team and customer familiarity with Agility for successful Agile projects.	L5	CO5	[8M]

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Model

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Planning Doum

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Sreenidhi Institute of Science and Technology

(An Autonomous Institution)

Regulations: A22

Code No:9ZC23

Date:19-Dec-2025 (AN)

B.Tech IV-Year I-Semester External Examination, December-2025 (Regular)

ADVANCED ENTREPRENEURSHIP (OE-II) (MECH,CSE,IT,CS,AIML,DS & IOT)

Time: 3 Hours

Max.Marks:60

Note: *a) No additional answer sheets will be provided.*
b) All sub-parts of a question must be answered at one place only, otherwise it will not be valued.
c) Missing data can be assumed suitably.

Bloom's Cognitive Levels of Learning (BCLL)

Remember	L1	Apply	L3	Evaluate	L5
Understand	L2	Analyze	L4	Create	L6

Part - A

Max.Marks: 6x2=12

ANSWER ALL QUESTIONS, EACH QUESTION CARRIES 2 MARKS.

		BCLL	CO(s)	Marks
1	List any two types of business models used in startups.	L1	CO1	[2M]
2	Explain the term "financial forecasting" in business planning.	L2	CO2	[2M]
3	Define the term "A-Team" in the context of startup hiring.	L1	CO3	[2M]
4	Define brand positioning and its importance for startups.	L1	CO4	[2M]
5	What do you mean by Customer Acquisition Cost (CAC)?	L1	CO5	[2M]
6	Who are startup mentors and why are they important.	L2	CO6	[2M]

Part – B

Max.Marks: 6x8=48

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 8 MARKS.

		BCLL	CO(s)	Marks
7.	Define "Product manager" and explain the role and significance of product manager.	L2	CO1	[8M]
	OR			
8	Describe the key factors that must be evaluated before refining a product or service.	L2	CO1	[8M]
9.	Illustrate the steps involved in preparing a financial plan for a new business. Explain how financial forecasting supports decision-making.	L4	CO2	[8M]
	OR			
10	Explain the customer life cycle for a growth-oriented business.	L2	CO2	[8M]
11	Explain the different strategies to attract talent and build a strong A-Team within a startup.	L2	CO3	[8M]
	OR			
12	Summarize the various stages of funding for startups.	L6	CO3	[8M]
13	Discuss about creating brand name and logo with an example.	L2	CO4	[8M]
	OR			
14	"Technology is a competitive advantage for startups." — Discuss this statement with reference to digital tools such as social media etc.	L5	CO4	[8M]
15	Discuss the key financial metrics (CAC, CLV, ARPU) used to measure startup performance.	L2	CO5	[8M]
	OR			
16	Explain the steps involved in trademark and company name registration.	L2	CO5	[8M]
17	Assess the process of selecting suitable mentors and advisors for a startup. What criteria should founders use when choosing them?	L4	CO6	[8M]
	OR			
18	Describe the steps involved in developing a final capstone project, from idea selection to presentation.	L1	CO6	[8M]

Sales / Total cost   total revenue – CAC   Revenue / user

Start Expan Exit

Launch Acquire develop retention

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Regulations: A22

Sreenidhi Institute of Science and Technology

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Code No:9EC09 **Date:13-Dec-2025 (AN)**

B.Tech IV-Year I-Semester External Examination,December-2025(Regular)

CRYPTOGRAPHY AND NETWORK SECURITY (CSE & IT)

Time: 3 Hours **Max.Marks:60**

Note: *a) No additional answer sheets will be provided.*
b) All sub-parts of a question must be answered at one place only, otherwise it will not be valued.
c) Missing data can be assumed suitably.

Bloom's Cognitive Levels of Learning (BCLL)

Remember	L1	Apply	L3	Evaluate	L5
Understand	L2	Analyze	L4	Create	L6

Part - A

Max.Marks: 6x2=12

ANSWER ALL QUESTIONS, EACH QUESTION CARRIES 2 MARKS.

		BCLL	CO(s)	Marks
1	Define Interruption and Interception attacks.	L1	CO1	[2M]
2	Define symmetric key encryption.	L1	CO2	[2M]
3	What is the purpose of Certificate Authority?	L2	CO3	[2M]
4	What is the purpose of Authentication Header (AH)?	L1	CO4	[2M]
5	Define a computer virus.	L1	CO5	[2M]
6	What is the purpose of a firewall in network security?	L1	CO6	[2M]

Part – B

Max.Marks: 6x8=48

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 8 MARKS.

		BCLL	CO(s)	Marks
7.	Explain different types of security attacks: Interruption, Interception, Modification and Fabrication with examples.	L2	CO1	[8M]
	OR			
8	Provide the architecture and need for Internet standards and RFCs in network security.	L4	CO1	[8M]
9.	Compare DES, TDES, 3DES & AES algorithms.	L4	CO2	[8M]
	OR			
10	Evaluate the strengths and limitations of symmetric encryption for large-scale systems.	L5	CO2	[8M]
11	Analyze with necessary mathematical steps of Diffie–Hellman key exchange.	L2	CO3	[8M]
	OR			
12	Interpret PGP architecture and its key management.	L4	CO3	[8M]
13	Compare and contrast AH and ESP based on functionality and security services.	L4	CO4	[8M]
	OR			
14	Provide the overview and architecture of IP Security.	L3	CO4	[8M]
15	Appraise the requirements of web security in detail.	L3	CO5	[8M]
	OR			
16	Categorize the types of intruders and intrusion detection mechanisms.	L4	CO5	[8M]
17	Outline the design principles of firewalls and discuss their role in network protection.	L3	CO6	[8M]
	OR			
18	Design security architecture for an organization using firewalls, trusted systems, and IDS.	L6	CO6	[8M]

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Regulations: A22

Sreenidhi Institute of Science and Technology
(An Autonomous Institution)

Code No:9EC08 **Date:15-Dec-2025 (AN)**

B.Tech IV-Year I-Semester External Examination,December-2025(Regular)
LINUX PROGRAMMING (CSE & IT)

Time: 3 Hours **Max.Marks:60**

Note: a) *No additional answer sheets will be provided.*
b) *All sub-parts of a question must be answered at one place only, otherwise it will not be valued.*
c) *Missing data can be assumed suitably.*

Bloom's Cognitive Levels of Learning (BCLL)

Remember	L1	Apply	L3	Evaluate	L5
Understand	L2	Analyze	L4	Create	L6

Part - A

Max.Marks: 6x2=12

ANSWER ALL QUESTIONS, EACH QUESTION CARRIES 2 MARKS.

		BCLL	CO(s)	Marks
1	Write any two Linux file handling utilities.	L1	CO1	[2M]
2	Define input and output redirection in shell scripting.	L2	CO2	[2M]
3	Write bout kernel support for files in Linux.	L1	CO3	[2M]
4	What is a zombie process?	L2	CO3	[2M]
5	Define pipe and mention one use.	L1	CO4	[2M]
6	What is a semaphore?	L2	CO4	[2M]

Part – B

Max.Marks: 6x8=48

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 8 MARKS.

		BCLL	CO(s)	Marks
7.	Explain commands related to Linux filters, file permissions and backup utilities with examples. Ciop tar	L2	CO1	[8M]
	OR			
8	Develop a command sequence using awk to extract usernames and home directories from /etc/passwd and explain the execution.	L3	CO1	[8M]
9.	Write a shell script to automate backup of a folder and schedule it using crontab. Explain each step.	L4	CO2	[8M]
	OR			
10	Discuss shell variables, quoting mechanisms, and flow control statements in Bash with examples.	L3	CO2	[8M]
11	Explain the Linux file system structure, standard I/O, and kernel support for file management.	L2	CO3	[8M]
	OR			
12	Demonstrate Linux file error handling methods and system calls: open(), read(), write(), and close() using C Programming Language.	L5	CO3	[8M]
13	Explain process creation, termination, and process states using suitable diagrams.	L2	CO3	[8M]
	OR			
14	Write about the following a. Process API b. Signals and Signal Generation c. Process abort and sleep function.	L5	CO3	[8M]
15	msgget() msgctl() Explain message queues and shared memory mechanisms with suitable UNIX system V APIs. shmat dt	L3	CO4	[8M]
	OR			
16	Develop an IPC-based client-server program using pipes for sending and receiving messages. Analyze its functionality.	L4	CO4	[8M]

17	Design a producer–consumer program using shared memory and semaphores to synchronize operations.	L5	CO4	[8M]
	OR			
18	Explain how semaphores avoid race conditions, and also how multiple processes access shared resource scenario.	L3	CO4	[8M]

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Sreenidhi Institute of Science and Technology
(An Autonomous Institution)

Regulations: A22

Code No:9EC07 **Date:22-Dec-2025 (AN)**

B.Tech IV-Year I-Semester External Examination, December-2025 (Regular)

AUTOMATA THEORY AND COMPILER DESIGN (CSE & IT)

Time: 3 Hours **Max.Marks:60**

Note: a) *No additional answer sheets will be provided.*
b) *All sub-parts of a question must be answered at one place only, otherwise it will not be valued.*
c) *Missing data can be assumed suitably.*

Bloom's Cognitive Levels of Learning (BCLL)

Remember	L1	Apply	L3	Evaluate	L5
Understand	L2	Analyze	L4	Create	L6

Part - A   Max.Marks: 6x2=12

ANSWER ALL QUESTIONS, EACH QUESTION CARRIES 2 MARKS.

		BCLL	CO(s)	Marks
1	Define an alphabet and a string. Give two examples of each.	L1	CO1	[2M]
2	Construct a finite automaton for the regular expression (0+1)1∗0.	L3	CO2	[2M]
3	List the main components of a Turing Machine model.	L2	CO3	[2M]
4	What is top-down parsing?	L1	CO4	[2M]
5	What type of relations does an operator-precedence parser use between terminals?	L1	CO5	[2M]
6	What is an S-attributed definition in the context of Syntax Directed Translation?	L2	CO6	[2M]

Part – B   Max.Marks: 6x8=48

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 8 MARKS.

		BCLL	CO(s)	Marks
7.	Explain any three operations on languages (such as union, concatenation, or Kleene star) with examples.	L1	CO1	[8M]
	OR			
8	Convert the following NFA to an equivalent DFA and determine the total number of DFA states: States = {q0, q1, q2} Alphabet = {a,b} δ(q0, a) = {q0} δ(q0, b) = {q0} δ(q0, a) = {q1} δ(q1, b) = {q2}	L3	CO2	[8M]
9.	Simplify the following grammar by removing ε-productions, unit productions, and useless symbols: $S \rightarrow AB \mid a$ $A \rightarrow \varepsilon \mid a$ $B \rightarrow b$	L4	CO1	[8M]
	OR			
10	Construct a derivation tree for the string aaabbb generated by the grammar $S \rightarrow aSb \mid ab$.	L4	CO2	[8M]
11	Design a non-deterministic pushdown automaton (NPDA) to accept the language $L = \{ a^n b^n c^m \mid n, m \geq 1 \}$ by empty stack, and explain why determinism is not possible for this language.	L5	CO1	[8M]

=) INPUT TAPE

=) FINITE MACHINE

OR

12 Design a turing machine to accept the language $L = \{ a^n b^n c^n \mid n>=1\}$ L4 CO3 [8M]

13 For the grammar: S → aSbb | ab L3 CO3 [8M]
Write the recursive-descent procedures and show a case where backtracking is unavoidable. Explain how this affects efficiency.

OR

14 Discuss phases of compiler with example. L5 CO4 [8M]

15 Given the grammar: S → Aa | bAc | dc | bda, A → d. L4 CO4 [8M]
Construct all LR(0) item sets. How many LR(0) states are generated in total? 12

OR

16 Construct all SLR(1) items for grammer: S → AB , A → aA/b, B->aB/b. L5 CO5 [8M]
and parse the string abab.

17 Generate 3-address code for the expression: a = b * -c + d / e. L3 CO4 [8M]
How many temporary variables are needed?

OR

18 For a nested expression: x = (a + b) * (c + d) - e / f, L4 CO6 [8M]
Determine the number of 3-address instructions required.

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